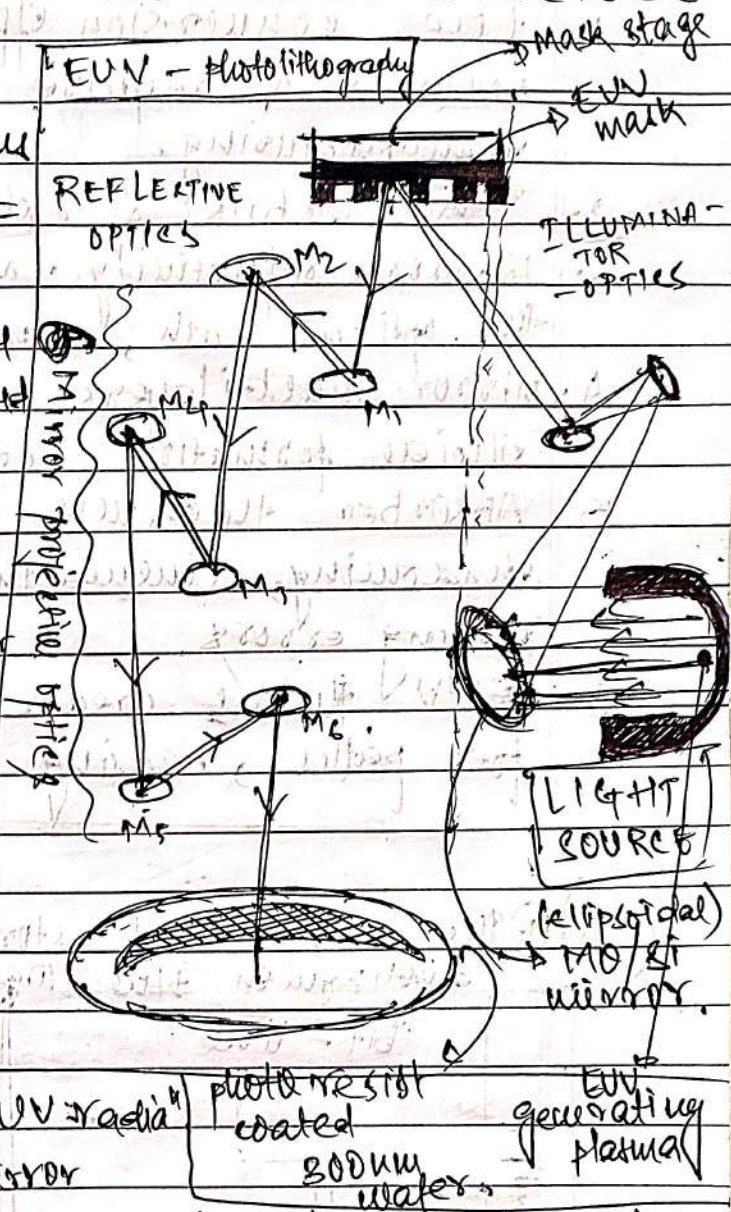


Nano → HW/3.

09.503: Nanoelectronics.

1. EUV lithography: performs projection imaging at a $\lambda = 13.5 \text{ nm}$ far below optical lithography limits. Because EUV light is strongly absorbed by air & all refractive materials, the entire exposure operates in high vacuum & uses fully reflective multilayer optics.



1. Laser produced plasma (LPP) generates radiation using high powered CO_2 laser (140-60kW) which strikes Cu droplets, vaporizes them to form intense plasma emitting EUV radiation.
2. Multilayer Mo/Si collector mirror captures & focus this light to intermediate focus (IF). Hydrogen buffer gas mitigates Cu debris via $\text{Cu} + 4\text{H} \rightarrow \text{CuH}_4$ (rxn).

3. The focussed light is directed to scanner optics, where multilayer Bragg mirrors form ILLUMINATION & PROJECTION optics. (70% theoretical peak reflectance near 13.5nm)

4. Imaging uses a REFLECTIVE MASK consisting of 140 Mo/Si bilayers capped with Ru & topped with metal absorber (typically TaBN) & often protected by a pellicle to reduce particle-induced effects.

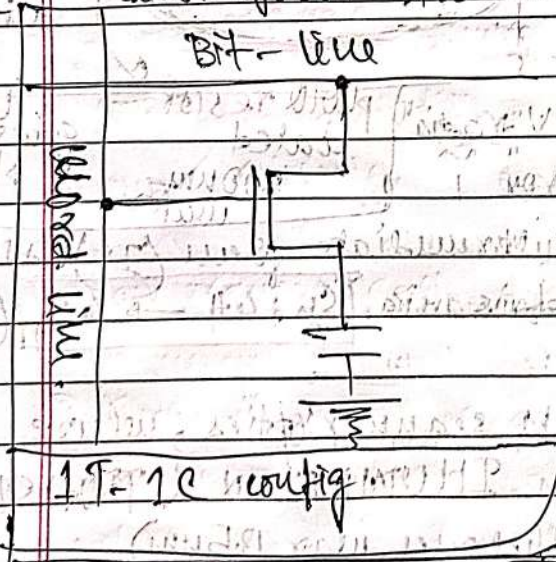
5. The wafer is coated with EUV sensitive photoresist. The demagnified image (4x reduce) is projected onto wafer through scanning exposure in vacuum.

Date ___/___/___

CHALLENGES:-

1. Low conversion efficiency (MBY) of EUV sources & require $>250W$ for high volume manufacturing.
2. Tm debris & collector mirror contamination.
3. Mirror reflectivity: $\sim 70\%$ reflectivity & $\sim 10^4$ mirrors, in the optical path, cumulative transmission is low.
4. Mirror multilayers suffer from interface roughness, etch formation, contamination & oxidation.
5. Absorber thickness AND off axis illumination causes shadowing, linewidth variation and orientation dependent imaging errors.
6. EUV photons create secondary e^- with long mean-free path, causing line-edge roughness.

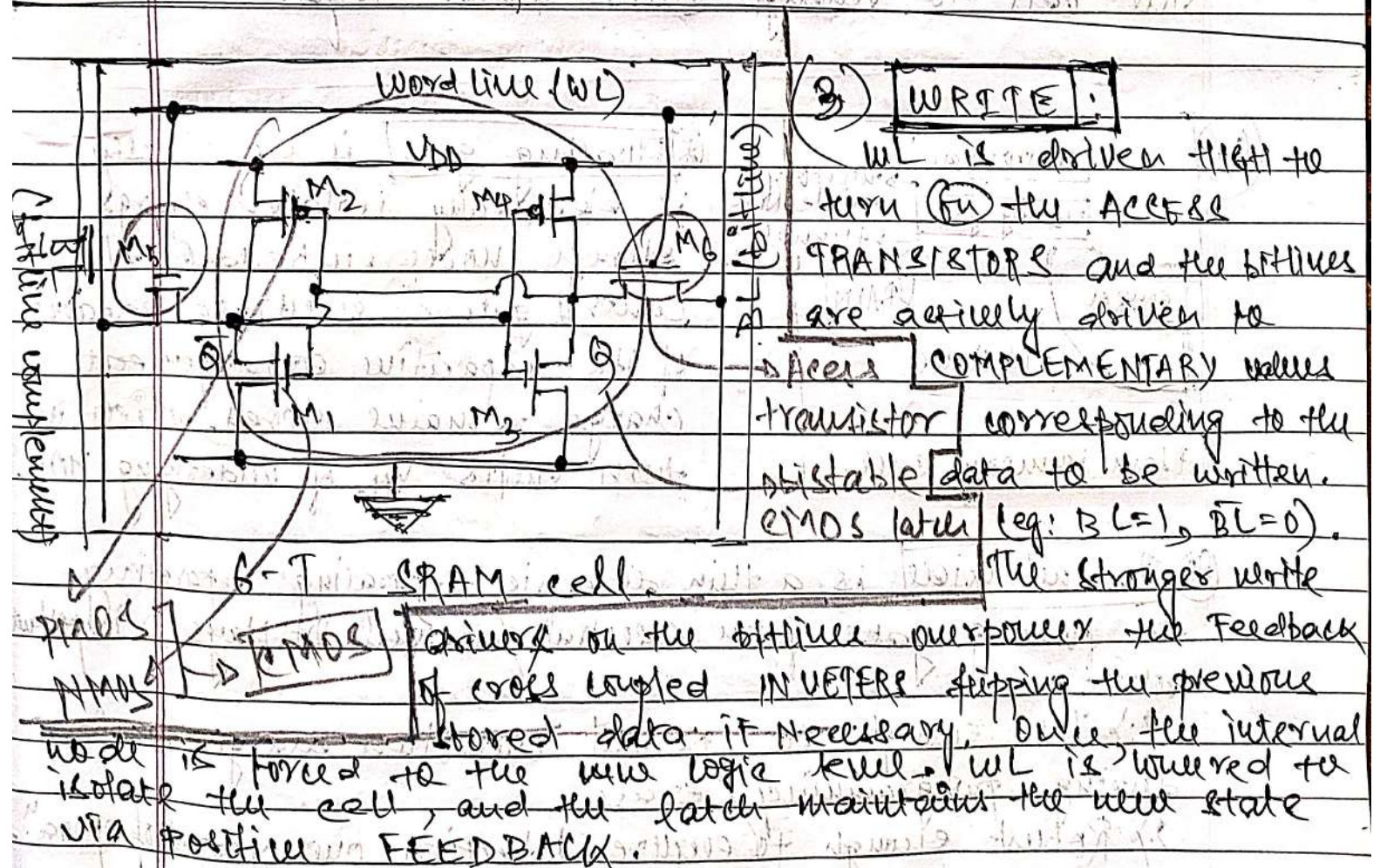
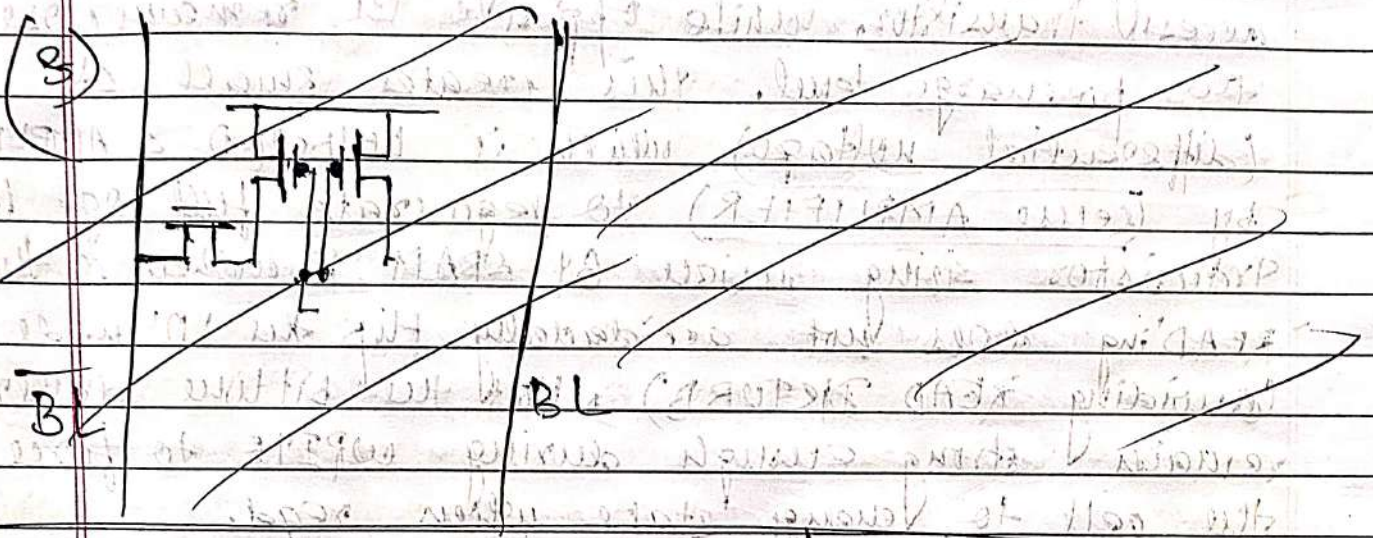
(*) ① The 1T1C DRAM cell stores information as charge. Ability to distinguish the logic 0 & logic 1 is governed by the stored charge $Q = C_{cell} \cdot V_{cell}$. Larger capacitance improves **SIGNAL MARGIN** at the sense amplifier. During READ operation: $\Delta V_{min} \propto \frac{Q}{C_{line} + C_{cell}}$. If C_{cell} becomes too small, the voltage change is comparable to noise & parasitic variations leading to read disturb errors & mis sensing.



② Leakage current & RETENTION dependent on cell capacitance. $t_{ret} \sim \frac{C_{cell} \cdot \Delta V_{min}}{I_{leak}}$ (ΔV_{min} : min detectable charge loss). Thus, higher capacitance improves retention time.

③ Capacitor leakage (self/tunnel/dielectric leakage) must remain extremely low ($< 10^{-4} A/cm^2$). Thin dielectrics

increase E field stress & leakage, therefore DRAM capacitor's ability to balance high capacitance low leakage directly affects DRAM reliability, refresh rate, noise margin and scalability.

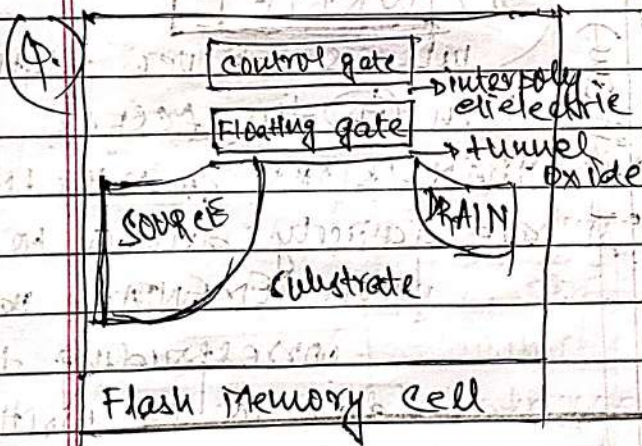


READ:

The cell must be disturbed minimally. Both bitlines

Date ___/___/___

are typically precharged to a mid/high level (eg: $V_{DD}/2$). A wordline (WL) is raised to turn on ACCESS transistors. This connects the two internal storage nodes to BL & $\overline{BL}$. Node storing '0' pulls bitline slightly downward through access transistor, while opposite BL remains close to the precharge level. This creates small ΔV (differential voltage) which is DETECTED & AMPLIFIED by (SENSE AMPLIFIER) to regenerate full logic level. Transistor sizing inside 6T1R1 is chosen so that READING does not accidentally flip the '0' node (avoiding READ DISTURB), but the bitline drivers remain strong enough during WRITE to force the cell to change state when reqd.



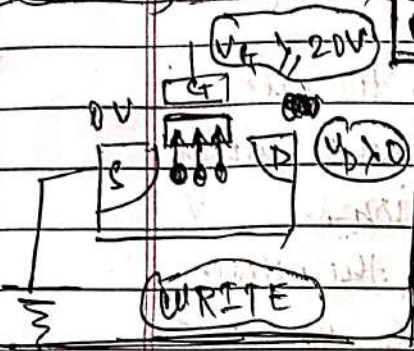
Floating gate is electrically isolated charge storage node embedded b/w control gate & substrate. Because of its capacitive arrangement, charge remains stored, which in turn shuts V_{th} of underlying MOS.

Tunnel oxide is a thin dielectric separating the floating gate from substrate/channel. It plays 2 important roles:

1. Thin enough to permit controlled charge transfer during programming/erase.
2. Robust enough to ensure retention over years of use.

Thickness & Quality determine reliability & endurance of MEMORY cell.

Date

WRITE / program :

A strong vertical field is applied on $V_G (V_G > V_D) \& (V_D > 0)$ [write] requires Hot carrier injection into the floating gate.
 Lateral (reqd): $V_{\text{Lateral}} = V_G - V_D$
 Vertical: $E_{\text{Vertical}} = \frac{V_G - V_D}{t_{\text{ox}}}$

WRITE

They surmount the Si-SiO₂ energy barrier & tunnel using FN (Fowler-Nordheim) tunneling.

Then accumulated charge (-ve) raises the cell's threshold voltage, so at a given read gate bias the channel does not invert & cell conducts essentially no current.

READ: control gate is given moderate voltage & drain current is sensed. [Floating gate $\rightarrow$ NO charge $\rightarrow$ low V_t transistor ON $\rightarrow$ Drain current (High) $\rightarrow$ '1'], but then [Floating gate $\rightarrow$ stored e^- $\rightarrow$ high V_t $\rightarrow$ transistor OFF $\rightarrow$ Drain current near 0 $\rightarrow$ '0'].

(5) ① At moderate / ambient vacuum, gas molecules continuously adsorb & desorb from sample surface, forming contaminants that disrupt / obscure intrinsic atomic / electronic struct.

② Characteristic time for contaminant monolayer formaⁿ scales inversely with pressure, so UHV extends this time allowing stable measurements.

③ E beams / tunneling currents used in these techniques modifies adsorbed layers. UHV reduces such parasitic interactions.

(6) (a) Contact AFM (atomic force microscopy)

- ~~AFM~~ [contact mode] $\rightarrow$ tip physically touches the surface and is scanned while maintaining a nearly constant cantilever deflection. Step heights, roughness, grain structure, the whole topography can be characterized @ nanometer scale.
- Since tip is in mechanical contact, one can even measure adhesion & stiffness.
- Contact AFM particularly useful in evaluating uniformity of gate oxides, tunnel oxides & high- k layers, since small surface features can correlate with underlying thickness variations or process defects.

(b) In ~~contact~~ non-contact AFM, the cantilever is oscillated near its resonant frequency ($f_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$) (applied driving frequency $f \approx f_0$).

Due to this resonance, any small change in frequency f gets reflected & amplified. These small changes are caused by the tip-sample interaction forces ($\frac{\partial F}{\partial z}$). These attractive/repulsive forces modify $k \rightarrow k_{eff} \rightarrow k_{eff} = k + \frac{\partial F}{\partial z}$.

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad ; \quad f' = \frac{1}{2\pi} \sqrt{\frac{k_{eff}}{m}} \quad ; \quad \Delta f = f' - f_0$$

$$\Delta f = \left(\frac{\partial f}{\partial k} \cdot \Delta k \right) = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \cdot \frac{1}{2k} \cdot \frac{\partial F}{\partial z} \cdot \Delta k \approx \frac{1}{4\pi} \sqrt{\frac{k}{m}} \cdot \frac{\partial F}{\partial z} \cdot \Delta k$$

Date ___/___/___

~~$$\Delta f = \frac{1}{4\pi\sqrt{m}} \left(\frac{\partial^2 F}{\partial z^2} \right) \left(\frac{\partial K}{\partial z} \right) + \frac{\partial F}{\partial z} \cdot \frac{\partial^2 K}{\partial z^2}$$

$$\Rightarrow \Delta f \approx \frac{f_0}{2K} \frac{\partial F}{\partial z}$$~~

$$K_{eff} = K + \frac{\partial F}{\partial z} \Rightarrow \Delta K = \left[\frac{\partial F}{\partial z} \right] = (K_{eff} - K)$$

$$\frac{\partial f}{\partial K} = \frac{d}{dK} \left(\frac{f_0}{2\pi\sqrt{m}} \right) = \frac{d}{dK} \left(\frac{f_0}{2\pi\sqrt{m}} \cdot \frac{1}{\sqrt{K}} \right) = \frac{f_0}{2\pi\sqrt{m}} \cdot \frac{-1}{2(K)^{3/2}} = -\frac{f_0}{2K} \frac{\partial F}{\partial z}$$

$$\Delta f = \frac{f_0}{2K} \frac{\partial F}{\partial z}$$

→ This shift in freq is detected by the feedback loop which adjusts the tip-sample distance to revert back to 'f'. This 'regula' happens constantly, and this scans the entire topography of the surface.

- Reduces tip wear & sample damage.
- Can probe soft / loosely bound surface.
- Can probe long range Van Der Waals forces.

Q1) Applications of STM:

- ① Atomic scale surface imaging using the exponential tunnelling current sensitivity to tip-sample distance.
- ② Mapping local electronic structure because tunnelling spectroscopy

Date ____ / ____ / ____

- current reflects the local density of states (DOS)
- ③ Bias-dependent imaging to distinguish filled vs empty electronic states.
 - ④ Characterisation of defects, adsorbates, reconstruction AND semiconductor surface states with Atomic precision.

CHALLENGES

- ① Requires conductive / ~~se~~ samples, cannot image true insulators.
- ② Image ~~contrast~~ contrast mixes topography & LPOs, making interpretation nontrivial.
- ③ Strong sensitivity to vibration, thermal drift, contamination often requiring UHV operation.
- ④ Tip condition is critical; blunt / contaminated tips distort atomic features.
- ⑤ Limited scan speed & small scanning area due to stability constraints.

(8) In SEM, a focused e^- beam scans across the sample surface and produces various signals as it interacts with the material. Secondary electrons are low-energy electrons ($\sim 50\text{eV}$) emitted from near the surface as the incident beam transfers energy via inelastic scattering. Because their escape depth is only a few nanometers, the SE signal is highly surface-sensitive and provides detailed topographical contrast: edges, steps, and fine surface features appear bright due to enhanced SE emission.

Backscattered Electrons (BSEs) are incident beam e^- that undergo [primary elastic] large-angle scattering and are reflected back out of the sample with ~~relatively~~ relatively high energy. The BSE yield depends strongly on atomic number Z : higher- Z regions produce more BSEs and appear brighter in BSE imaging. As a result, BSE contrast is more sensitive to composition and phase differences than to fine topography. By detecting SE & BSE separately or in combination, SEM can provide complementary information: SE images emphasize surface morphology, while BSE images highlight compositional variations, or buried features influencing scattering.

(9) In TEM, high energy with relativistic de Broglie wavelengths and hence allow atomic scale resolution.

① As the incident e^- wave passes through the thin specimen, it experiences the specimen's electrostatic potential and acquires a position-^(crystal's)dependent phase shift of the form $\psi_{\text{exit}} = \psi_0 \exp(i\sigma \int V dz)$, which encodes BOTH thickness and structural information.

② Crystalline lattices then diffract the beam according to the Bragg condition $\vec{k}_{\text{out}} = \vec{k}_{\text{in}} + \vec{g}$ [$\vec{g} \rightarrow$ lattice $\vec{h}$], producing distinct diffracted components.

③ The objective lens performs a Fourier transform of the exit wave and recombines the transmitted & diffracted beams. The interference of these components, filtered by the lens transfer characteristics, forms the real-space image. As a result TEM contrast reflects the combined effects of phase modulaⁿ, diffracⁿ & mass-thickness.